Fiscal Year:	FY 2004	**Task Last Updated:**	FY 03/29/2006
PI Name:	McPherson, Alexander Ph.D.		
Project Title:	Enhanced Dewar Program		
Division Name:	Physical Sciences		
Program/Discipline:	BIOTECHNOLOGY		
Program/Discipline--Element/Subdiscipline:	BIOTECHNOLOGY--Structural biology		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	amcphers@uci.edu	**Fax:**	FY 949-824-1954
PI Organization Type:	UNIVERSITY	**Phone:**	949-824-1931
Organization Name:	University of California, Irvine		
PI Address 1:	Department of Molecular Biology and Biochemistry		
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PI Web Page:	http://www.faculty.uci.edu/		
City:	Irvine	**State:**	CA
Zip Code:	92697-3900	**Congressional District:**	48
Comments:			
Project Type:	Flight	**Solicitation / Funding Source:**	OTHER
Start Date:	03/01/1995	**End Date:**	08/01/2004
No. of Post Docs:	0	**No. of PhD Degrees:**	
No. of PhD Candidates:	0	**No. of Master' Degrees:**	
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA MSFC
Contact Monitor:	Ciszak, Ewa	**Contact Phone:**	256 544-5531
Contact Email:	ewa.ciszak@msfc.nasa.gov		
Flight Program:	Shuttle		
Flight Assignment:	STS-106, STS-98, STS-104, STS-105, STS-110 NOTE: per information in TB rpt, end date chgd to 8/1/2004 (original end date was 12/01/2004)--jp 2/2009		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Jenkins, G. (University of California Irvine)		
Grant/Contract No.:	NAS8-00019		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	Initially, a relatively simple system composed of a Dewar flask filled with small crystallization samples will be frozen on Earth and allowed to passively thaw in microgravity. Samples will be housed in the Passive GN2 Freezer, a Dewar previously used in other microgravity applications. In addition to the crystallization experiments, plans are outlined for examination of crystals by standard light microscopy and also electron microscopy and atomic force microscopy, and detailed documentation of results, structure determination and high resolution refinement, and detailed analyses of structures obtained from crystals grown on Earth and in microgravity. The objective of the research is to develop a microgravity suitable system for screening protein crystallization conditions using the batch and liquid-liquid crystallization method. These experiments represent an exciting and important new opportunity for protein crystal growth studies in microgravity. The proteins and protein solutions that will tolerate the freeze-thaw mechanism used to initiate these experiments will increase as the trend continues away from exclusion of glycerol from protein solutions used in crystallization. Shuttle experiments, furthermore, suggest that most orbiting platforms have some variations in the acceleration environment, and some protein crystallizations are more tolerant of these variations than others. Many of the difficulties that have affected most previous microgravity experiments can be circumvented in the Dewar system and long duration flights making possible crystallization experiments that cover much longer equilibration times.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	
Task Progress:	On June 28, 2004, UCI received a contractual letter requesting an early completion of the contract. A final report was developed and submitted to NASA. The EGN completed five microgravity missions and involved 1200 teachers and students in the preparation of flight samples. Successful missions completed were STS-106, STS-98, STS-104, STS-105, and STS-110. There were 1077 samples loaded of which 728 were flown. Approximately 75,000 students participated in the classroom activity and competed for the opportunity to prepare these flight samples. The final report was completed on August 8, 2004 and 15 copies were delivered to the COTR. The main body of this report included a summary of the project. Copies of data requirements that have been previously submitted to NASA were also included as attachments for the COTR's file copy. A final New Technology/Invention Report was submitted on September 13, 2004. This report documents there were no reportable items developed on the contract.
Bibliography Type:	Description: (Last Updated: 02/27/2004)